

84 people still inside. Room 314 has people who cannot use the stairs on their own.

The firefighter view shows the safest way in to reach them.

SENSE AI

le still inside

Time to get everyone out

1m 20s

if they follow the plan

Are people following it?

—

nobody is following yet

Level 3 — simplified floor plan



Who is still inside?

The one question a fire alarm has never been able to answer.

1. IT COUNTS PEOPLE

HKTech300 · 21st Cohort | Suvum · Sartavev · Yerdos · Rauan

2. IT SHARES OUT THE EXITS

3. IT CHECKS AND CORRECTS

WANG FUK COURT · TAI PO · 26 NOVEMBER 2025

168

people died in a Hong Kong housing estate, 40 minutes
from this room.

Hong Kong's deadliest fire since 1948. 79 iniured. The fire burned for 44 hours. Source: Wikipedia. CNN. SCMP.

WHAT WENT WRONG THAT NIGHT

Three failures. Only one of them was the fire.

0 of 8

towers where a fire alarm actually sounded. Residents were warned by neighbours knocking on doors.

467

missing-person reports filed, because nobody could say who was still in the building.

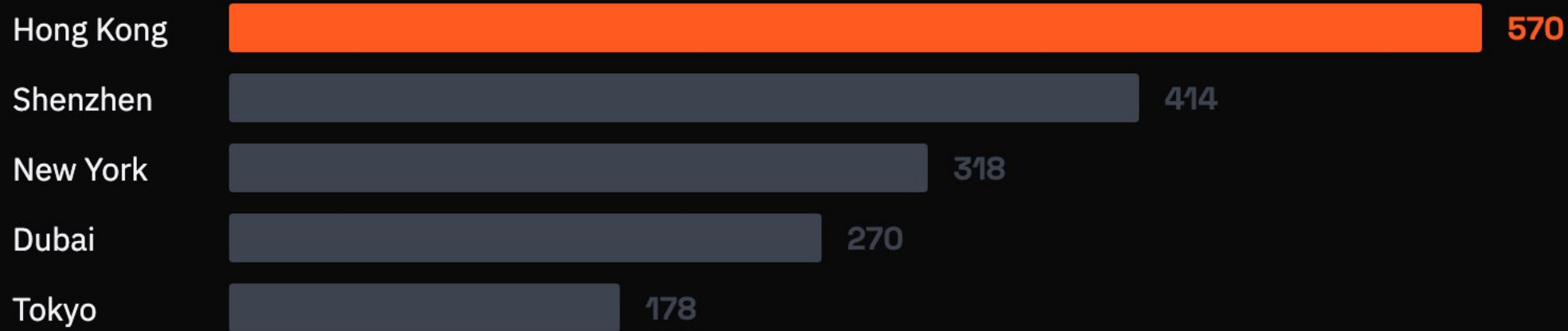
Most

of those who died were found inside their own flats. They never made it to a staircase.

We cannot fix an alarm that was never maintained. We can answer the second question: **who is still inside, and where.**

WHY HONG KONG

The most vertical city on earth.



Buildings over 150 metres. Hong Kong also has **more than 9,000** high-rise buildings in total, and over 1,000 above 100 metres.

THE PART NOBODY DESIGNS FOR

People do not run. That is the whole problem.

3 min

median delay before anyone in the World Trade Center began to move, after the alarm.

45 min

the far end of the range in high-rise studies. Some people never start at all.

Then they do the second thing

They walk to the staircase they came in by — not the nearest one, and not the emptiest one.

One core jams. The other two stand half empty. The building empties slower than it was designed to.

WHAT WE BUILT

It watches where people actually go — and moves them.



84

people counted, room by room

3

staircases, balanced by travel time

74

times the plan was rewritten, because people disobeyed it

THE TECHNOLOGY

Two sensors. Opposite failures.

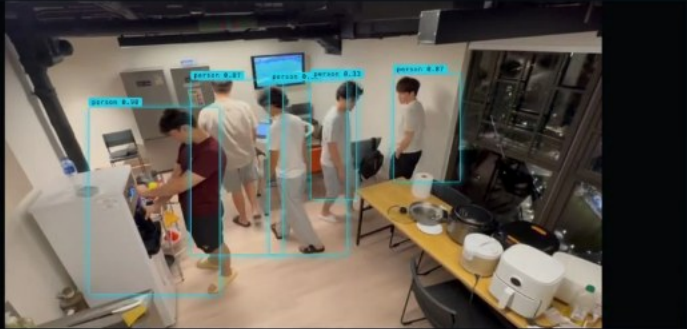
People in each room				about 52 people on this floor	Show empty rooms
Room	People	How sure we are	Who is telling us	Air	
West corridor	0-1	Fairly sure	Radar	Clear	
West-central corridor	0-1	Fairly sure	Radar	Clear	
Central corridor	0-1	Fairly sure	Radar	Clear	
East-central corridor	0-1	Fairly sure	Radar	Clear	
East corridor	0-1	Fairly sure	Radar	Clear	
Room 301 open-plan desks	4-5	Fairly sure	Radar	Clear	
Room 302 meeting room	4-6	Not sure	Radar	Clear	
Room 303 open-plan desks	3-4	Fairly sure	Radar	Clear	
Room 304 lab	2-4	Not sure	Radar	Clear	
Room 305 client and server room	0-2	Not sure	Radar	Clear	
Room 306 open-plan desks	3-4	Fairly sure	Radar	Clear	
Room 311 open-plan desks	2-3	Fairly sure	Radar	Clear	
Room 312 meeting room	2-4	Fairly sure	Radar	Clear	
Room 313 storeroom	0-2	Not sure	Radar	Clear	
Room 314 open-plan desks	7-9	Not sure	Radar	Clear	
Room 315 quiet room	1-2	Fairly sure	Radar	Clear	
Room 316 meeting room	1-2	Not sure	Radar	Clear	

Smoke blinds the camera. Radar keeps counting.

What the camera actually sees

Play the example Choose file sample.mp4 Use my own video

sample.mp4: 40 frames sampled over 10.0s - 190 detections



People it can see: 5 (peak: 7 in clip)

How sure it is: 76% (about what it found)

How much smoke: 0% (blotting the view)

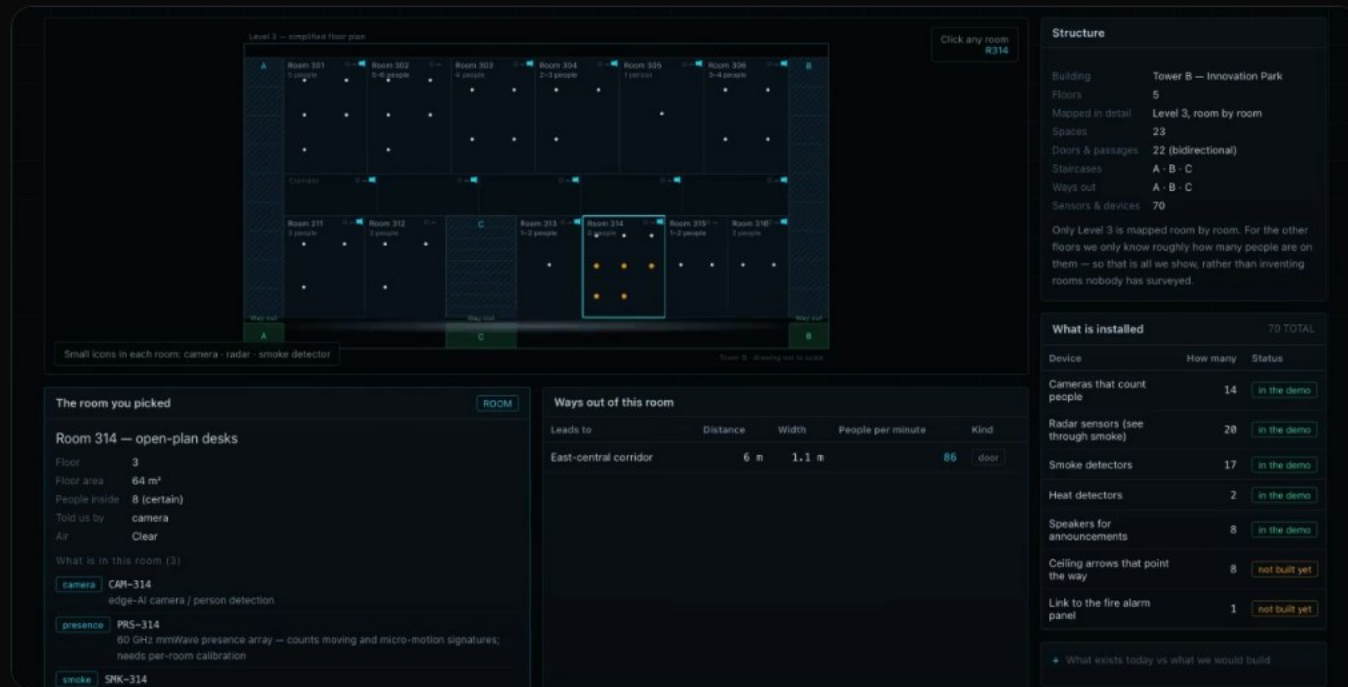
Fire or smoke?: none (specified in the picture)

Camera count usable: fused with the presence layer

Our own footage, live. Seven people, boxes drawn by the model.

WHAT ACTUALLY GETS INSTALLED

Four devices a floor. One model of the building.



Camera

counts people while the air is clear

Radar

keeps counting once it is not

Smoke & heat

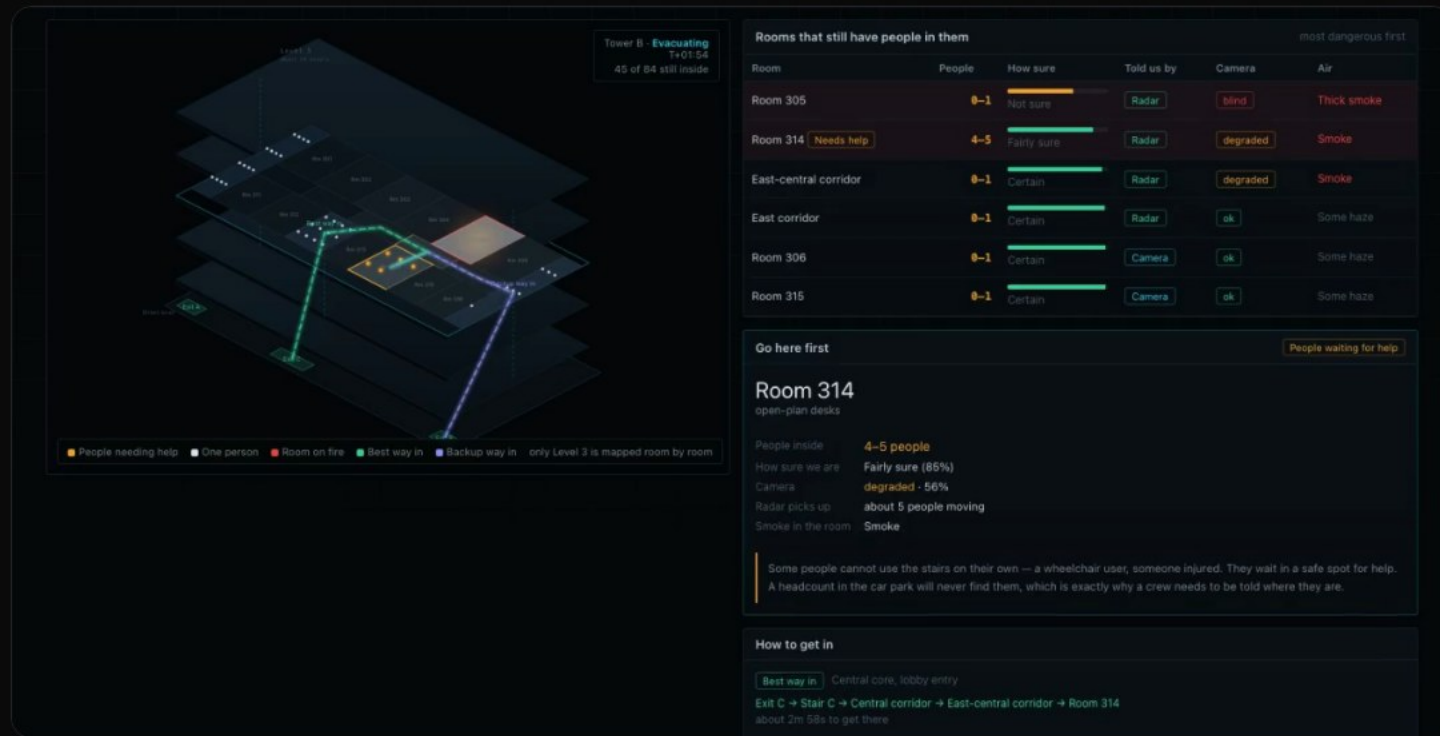
the existing panel, read by us

Speakers

a different instruction per zone

FOR THE CREW ARRIVING

467 missing-person reports. Or one screen.



Room 314

8 people · camera blind · radar holding

Plus a way **in** that avoids the crowd coming **out**.

MARKET

1,000 buildings tall enough to need this.

TAM

HK\$120M

Every Hong Kong building over 100 m, at HK\$120k a year.

SAM

HK\$68M

The 570 towers over 150 m, which already employ fire-safety staff.

SOM

HK\$2M

3% of those by year three — about 17 buildings.

We start with managing agents who run many towers: one relationship, one procurement conversation, many buildings.

COMPETITION

Everyone watches the building. Nobody watches the people.

↑ Knows where the people are

- Occupancy analytics
Density, VergeSense

- **SENSE AI**
Live people, live fire,
and a plan that corrects
itself

- Fire alarm panels
Siemens, Honeywell

- Evacuation simulators
Pathfinder, STEPS

- Adaptive exit signage

Knows where the fire is →

BUSINESS MODEL

Two ways in. Both recurring.

DIRECT · BUILDING OWNERS

HK\$120k

per building, every year
plus HK\$22k per floor to fit out

A 25-floor tower: HK\$550k once, then HK\$120k a year.

CHANNEL · FIRE-SAFETY FIRMS

HK\$45k

per building, every year
they keep the hardware margin

Lower price, far faster reach — they already hold the customer.

No revenue yet. We are testing whether one owner will buy, and one installer will resell.

TEAM

Four founders. We wrote the engine ourselves.

CEO

Bekarys Suyum

2nd year

Direction, and the conversations with building owners.

CTO

Sanzhar Sartayev

1st year

Wrote the routing, fusion and simulation engine. ISEF national champion.

CFO

Zhoraev Yerdos

1st year

Pricing, unit economics, the three-year plan.

COO

Khamza Rauan

2nd year

Pilots — sensors onto real floors, kept running.

The gap we know about: none of us is a certified fire engineer. That is our first hire.

THREE YEARS

Validate, repeat, scale.

YEAR 1 · VALIDATE

- Fire-safety advisor hired
- Two pilot buildings
- First paid pilot

HK\$0.2M

YEAR 2 · REPEAT

- 8 to 12 buildings live
- One installer reselling
- Alarm-panel integration

HK\$1.2M

YEAR 3 · SCALE

- 15 to 20 buildings
- Certification pathway open
- Second market

HK\$2.0M

A century of getting better at finding **fire**.

We think the next one is about finding the people.

SENSE AI is a working prototype, not a certified life-safety system.